

	SR UNIVERSITY	
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	Program: II - B. Tech (CS& AI)	Batch: 57
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	AI Assisted Coding-Assignment	Date: 18-09-2026

ASSIGNMENT-07

1. Question

Q1. Class with Conditional Logic – Employee Salary

Scenario:

An AI coding assistant is used to develop a simple employee salary management program.

Task:

Using AI-assisted code completion, write a Python class Employee that:

- Stores employee name and basic salary.
- Has a method calculate_bonus() that:
 - Returns 10% of salary if salary is ₹50,000 or above.
 - Returns 5% of salary otherwise.
- Has a method display_details() to display the employee name, salary, and bonus.
- Create at least two employee objects and test the class.

2. Prompt

Create a Python program using an Employee class with name and basic_salary. Calculate a **10% bonus if salary ≥ ₹50,000**, otherwise **5%**. Calculate and display the total salary. Create two employee objects and test both conditions.

3. Code

```
lab6.py > ...
1 class Employee:
2     def __init__(self, name, basic_salary):
3         self.name = name
4         self.basic_salary = basic_salary
5
6     def calculate_total_salary(self):
7         if self.basic_salary >= 50000:
8             bonus = self.basic_salary * 0.10
9         else:
10            bonus = self.basic_salary * 0.05
11
12            return self.basic_salary + bonus
13
14    def display_salary(self):
15        total_salary = self.calculate_total_salary()
16        print(f"Employee: {self.name}")
17        print(f"Basic Salary: ₹{self.basic_salary:,.2f}")
18        print(f"Total Salary: ₹{total_salary:,.2f}")
19
20
21    employee1 = Employee("Aisha", 60000)
22    employee2 = Employee("Rahul", 40000)
23
24    employee1.display_salary()
25    print()
26    employee2.display_salary()
```

4. Output

```
PS C:\My programs\AI Assisted Coding> python lab6.py
Employee: Aisha
Basic Salary: ₹60,000.00
Total Salary: ₹66,000.00

Employee: Rahul
Basic Salary: ₹40,000.00
Total Salary: ₹42,000.00
```

5. Observation

Aisha's salary is ₹60,000, so she receives a 10% bonus of ₹6,000. Total salary: ₹66,000.

Rahul's salary is ₹40,000, so he receives a 5% bonus of ₹2,000. Total salary: ₹42,000.

The program correctly tests both salary conditions.

1.QUESTION

Q2. Loop and Conditional Logic – Student Marks

Scenario:

A teacher wants to analyze the marks of students using an AI coding assistant.

Task:

Using AI-assisted code completion, write a Python program that:

- Accepts a list of student marks.
- Uses a loop to process each mark.
- Displays:
 - "Distinction" for marks ≥ 75
 - "Pass" for marks ≥ 40 and < 75
 - "Fail" for marks < 40
- Ignores invalid marks outside 0–100.
- Calculates and displays the number of students in each category.
- Test the program with a suitable list of marks.

2.PROMPT

create a python program which accepts lists of students marks and uses loop to process each mark

and if marks ≥ 75 -- distinction

marks ≥ 40 and < 75 -- pass

marks < 40 -- fail

if marks outside 0 and 100 invalids

create each test case and test all

3. CODE

```
def process_marks(student_marks):
    for marks in student_marks:
        if marks < 0 or marks > 100:
            result = "Invalid marks"
        elif marks >= 75:
            result = "Distinction"
        elif marks >= 40 and marks < 75:
            result = "Pass"
        else:
            result = "Fail"

        print(f"Marks: {marks} - {result}")

test_cases = [
    [85, 72, 35, 101, -5],
    [75, 40, 39, 100, 0],
]

for case_number, student_marks in enumerate(test_cases, start=1):
    print(f"Test case {case_number}:")
    process_marks(student_marks)
    print()
```

4.OUTPUT

```
• (.venv-1) PS C:\My programs\AI Assisted Coding> & "c:\My programs\AI Assisted Coding\.venv-1\Scripts\python.exe" "c:\My programs\AI Assisted Coding\Scripts\lab6.py"
Test case 1:
Marks: 85 - Distinction
Marks: 72 - Pass
Marks: 35 - Fail
Marks: 101 - Invalid marks
Marks: -5 - Invalid marks

Test case 2:
Marks: 75 - Distinction
Marks: 40 - Pass
Marks: 39 - Fail
Marks: 100 - Distinction
Marks: 0 - Fail
```

5.OBSERVATION

The program correctly processes every mark using a loop:

- 85, 75, and 100 → Distinction
- 72 and 40 → Pass
- 35, 39, and 0 → Fail
- 101 and -5 → Invalid marks

Question 3 :

Class + Loop – Product Inventory

Scenario: A small store wants to maintain product inventory using a Python program generated with AI assistance.

Task: Using AI-assisted code completion, create a class Product that:

- Stores product name, price, and quantity.
- Has a method `total_value()` that calculates price × quantity.
- Create a list of multiple Product objects.
- Use a loop to calculate the total inventory value.
- Display the products whose quantity is less than 5 as "Low Stock".
- Test the program with at least three products.

2. Prompt

Create a Python Product class with product name, price, and quantity. Calculate the total value of each product using price × quantity. Use a loop to find the total inventory value and display products with quantity less than 5 as Low Stock. Test with three products.

3.CODE

```

10 class Product:
11     def __init__(self, name, price, quantity):
12         self.name = name
13         self.price = price
14         self.quantity = quantity
15
16     def total_value(self):
17         return self.price * self.quantity
18
19
20 products = [
21     Product("Laptop", 75000, 3),
22     Product("Keyboard", 2500, 10),
23     Product("Mouse", 1200, 4),
24 ]
25
26 total_inventory_value = 0
27
28 for product in products:
29     product_value = product.total_value()
30     total_inventory_value += product_value
31     stock_status = " - Low Stock" if product.quantity < 5 else ""
32     print(f"{product.name}: ₹{product_value:,.2f}{stock_status}")

```

4.OUTPUT

```

• (.venv-1) PS C:\My programs\AI Assisted Coding> & "c:\My programs\AI Assisted Coding\.venv-
lab6.py"
Laptop: ₹225,000.00 - Low Stock
Keyboard: ₹25,000.00
Mouse: ₹4,800.00 - Low Stock
Total inventory value: ₹254,800.00

```

5.OBSERVATION

The program successfully processes all three products, calculates each product's total value, identifies low-stock products, and calculates the total inventory value of ₹254,800.00.

Q4. Class + Loop + Nested Conditions – Electricity Bill

Scenario:

An electricity board wants to calculate bills based on units consumed.

Task:

Using AI-assisted code completion, create a Python class ElectricityBill that:

- Stores the consumer name and units consumed.
- Calculates the bill using:
 - First 100 units → ₹2/unit
 - 101–200 units → ₹3/unit
 - Above 200 units → ₹5/unit
- Uses appropriate **loops and conditional statements** where required.
- Displays the consumer name, units, and total bill.
- Create at least three objects and display their bills.

Test the program with different unit values, including an edge value such as 100 or 200.

2.PROMPT

Create a Python ElectricityBill class to calculate electricity bills based on units consumed. Charge ₹2 per unit for the first 100 units, ₹3 per unit for units 101–200, and ₹5 per unit above 200. Display the consumer name, units, and total bill. Test with three consumers.and run program with different unit values, including an edge value such as 100 or 200

3.code

```

10 class ElectricityBill:
11     def __init__(self, consumer_name, units_consumed):
12         self.consumer_name = consumer_name
13         self.units_consumed = units_consumed
14
15     def calculate_bill(self):
16         units = self.units_consumed
17
18         if units <= 100:
19             return units * 2
20         elif units <= 200:
21             return (100 * 2) + ((units - 100) * 3)
22         else:
23             return (100 * 2) + (100 * 3) + ((units - 200) * 5)
24
25     def display_bill(self):
26         total_bill = self.calculate_bill()
27         print(f"Consumer Name: {self.consumer_name}")
28         print(f"Units Consumed: {self.units_consumed}")
29         print(f"Total Bill: ₹{total_bill:,.2f}")
30
31
32 consumers = [
33     ElectricityBill("Anita", 100),
34     ElectricityBill("Bharat", 200),
35     ElectricityBill("Charan", 250),
36 ]

```

4.OUTPUT

```

lab6.py"
Consumer Name: Anita
Units Consumed: 100
Total Bill: ₹200.00

Consumer Name: Bharat
Units Consumed: 200
Total Bill: ₹500.00

Consumer Name: Charan
Units Consumed: 250
Total Bill: ₹750.00

```

5.OBSERVATION

The program correctly applies progressive electricity rates:

- First 100 units at ₹2 per unit
- Units 101–200 at ₹3 per unit
- Units above 200 at ₹5 per unit